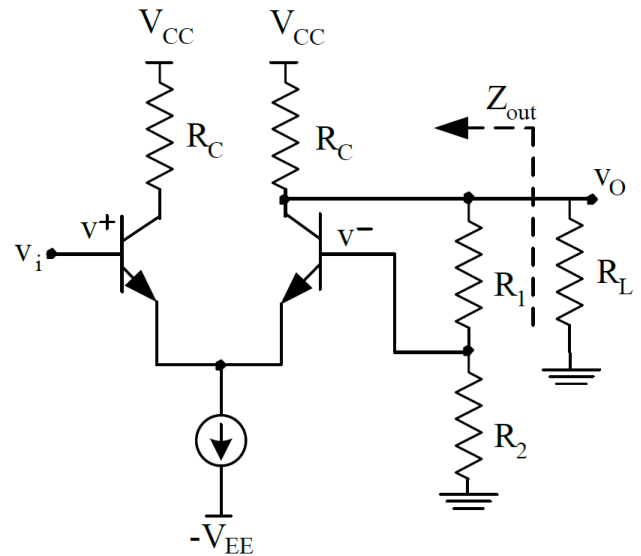


RE3 – Stability of a differential amplifier with feedback

Consider the feedback amplifier in figure consisting of an amplifier (A) and a feedback loop (β). Amplifier A is characterized by inputs V^+ and V^- , and output voltage V_o , with voltage gain $A = V_o/(V^+ - V^-) = 8000$, a differential input resistance $R_{id} = 10k\Omega$ and an output resistance $R_o = 5k\Omega$. Consider the load resistance $R_L = 100k\Omega$, and the feedback resistors $R_1 = 40k\Omega$ and $R_2 = 10k\Omega$.



- Identify the feedback topology used in the circuit and the stabilized parameter. Represent the circuit using the simplified electrical diagram of amplifier A , the loop network β and the load.
- Calculate the parameters of the matrices that describe A and β .
- Represent the circuit in terms of the blocks A' ($A_{charged}$) and β' and determine the values of A' and β' . If you didn't calculate the matrices in the previous point, consider the pair:

$$A = \begin{pmatrix} 15 \times 10^3 & -0,1 \\ -6000 & 2 \times 10^{-5} \end{pmatrix} \quad \beta = \begin{pmatrix} 3 \times 10^3 & 0,3 \\ -2000 & 2 \times 10^{-4} \end{pmatrix}$$

- Based on feedback theory, determine the voltage gain V_o/V_i and the output impedance Z_{out} of the feedback amplifier.
- Considering that the amplifier has a gain of $134dB$ and poles $f_{p1} = 1MHz$ and $f_{p2} = 10MHz$, and the loop β illustrated in figure, sketch the gain and phase characteristics of $A\beta$ and estimate the gain margin of the feedback amplifier. Dimension a dominant pole by displacing the first pole, in order to guarantee a phase margin of 45° .

Resolution

a)

Series-parallel or Voltage-Voltage topology.

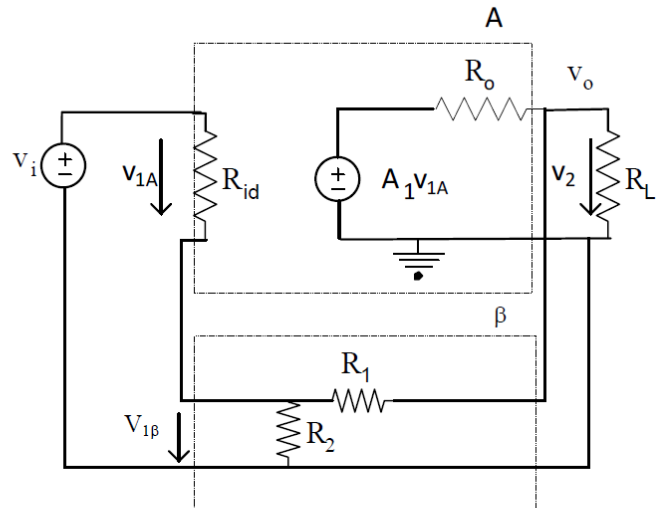
Stabilized parameter: Voltage gain (samples output voltage and compares input voltages).

b)

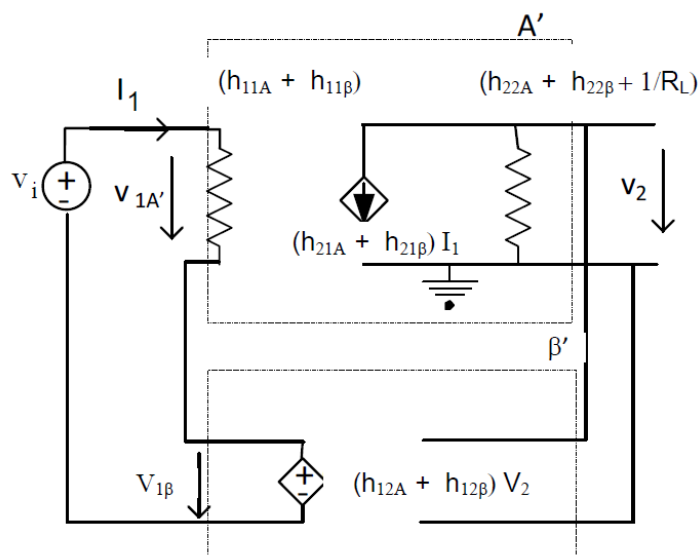
$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} R_{id} & 0 \\ -\frac{A_1 R_{id}}{R_0} & \frac{1}{R_0} \end{bmatrix} = \begin{bmatrix} 10k\Omega & 0 \\ -16k & 2 \times 10^{-4} S \end{bmatrix}$$

$$\beta = \begin{bmatrix} R_1 // R_2 & \frac{R_2}{R_1 + R_2} \\ -\frac{R_2}{R_1 + R_2} & \frac{1}{R_1 + R_2} \end{bmatrix} = \begin{bmatrix} 8k\Omega & 0,2 \\ -0,2 & 2 \times 10^{-5} S \end{bmatrix}$$



c)



$$A' = 3865 \quad \beta' = 0.2$$

$$A' = 1932 \quad \beta' = 0.2 \quad (\text{with the matrices given in the statement})$$

d)

$$Z_{of} = \frac{Z_{oA'}}{1 + A'\beta'} = 5.6\Omega, \quad Z_{oA'} = \frac{1}{h_{22A} + h_{22\beta} + 1/R_L}$$

$$Z_{out} = \frac{1}{\frac{1}{Z_{of}} - \frac{1}{R_L}} = 5.6\Omega$$

With the matrices given in the statement: $Z_{of} = 11.21\Omega$ and $Z_{out} = 11.21\Omega$

e)

$A_o = 134dB$, $\beta = 0.2 = -14dB$. Then at low frequency, $A\beta = 120dB$.

Before compensation, the phase margin is close to zero, although slightly positive, and the system is stable.

For the phase margin (PH) to be 45° with the first pole displaced, it has to be at $10Hz$.

